

Removing Monte Carlo noise using a Sobel operator and a guided image filter

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Abstract

Monte Carlo (MC) ray tracing is an effective method for generating photorealistic images, but it often suffers from noise artifacts at low sampling rates. This project proposes implementing a feature-based approach to remove Monte Carlo noise while preserving image details. The method involves the use of a Sobel operator to identify geometric structures and a guided image filter to remove noise from additional features such as depth and textures. The adaptive sampling and reconstruction approach then optimizes filter parameters to produce smooth images with minimal numerical error. Experimental results indicate that this method outperforms traditional approaches in both visual quality and computational efficiency. Referred to Research Paper: [\[LZZY18\]](#)

1 Introduction to Problem

Monte Carlo (MC) ray tracing is widely used in computer graphics for producing high-quality, photorealistic images by simulating the complex interactions of light with objects. However, the major drawback of MC rendering is the presence of noise artifacts, particularly at low sampling rates, due to the random nature of the sampling process. This noise can obscure important details and degrade the overall quality of the rendered image. Current noise reduction methods often struggle to balance noise removal with the preservation of fine details, leading to over-smoothed images or residual noise.

2 Approach to solve the problem

The proposed solution involves a novel feature-based adaptive rendering approach. The core idea is to use additional features such as depth, normals, and textures to guide the noise removal process. The Sobel operator is first applied to compute a gradient image that highlights geometric edges. This gradient image is then used to guide a filtering process that prefilters the noisy features using a guided image filter, which helps in preserving the fine details of the scene. Following this, an adaptive sampling strategy is employed where the spatial and feature parameters of the filters are optimized through a data-driven method. This allows for robust noise removal while maintaining high image fidelity. The approach is implemented as an extension of the PBRT (Physically Based Rendering Toolkit) to ensure efficient computation.

Step by Step Methodology:

1. Input and Feature Extraction:

- Start with a noisy Monte Carlo rendered image.
- Extract features like depth, normals, and textures from the scene.

2. Gradient Image Computation:

- Apply the Sobel operator to features to create gradient images.
- Construct a final gradient image to highlight structural details.

3. Feature Prefiltering:

- Use the gradient image to guide the filtering of noisy features with a guided image filter.

- This reduces noise in the features while preserving details.

4. Parameter Selection:

- Estimate the optimal spatial and feature parameters through parametric curve fitting and candidate filter evaluation.
- These parameters control the smoothing process to balance noise reduction and detail preservation.

5. Two-Step Smoothing:

- Estimate errors for each filter and create binary maps.
- Smooth these maps with guided filtering and compute a weighted average to produce the final filtered image.

6. Spike Noise Removal:

- Detect and down-weight spike pixels using gradient information.
- Apply a two-radius strategy to effectively remove spike noise without losing details.

7. Adaptive Sampling:

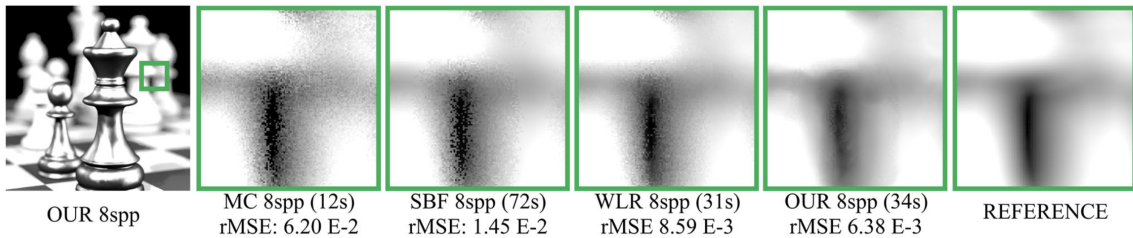
- Start with low sample counts and redistribute additional samples based on error estimation.
- Focus more samples on high-error regions to improve image quality.

8. Final Image Reconstruction:

- Use the optimized parameters to perform the final filtering.
- Produce a noise-reduced image that preserves key details.

3 Final Results and Summary

The implementation of this method is expected to result in significantly improved image quality in Monte Carlo ray-traced images, especially in scenarios involving complex geometries and lighting effects such as motion blur and depth of field. By leveraging the additional features and optimizing filter parameters adaptively, the method is able to reduce noise effectively without sacrificing image details. Preliminary results from the implementation show that this approach outperforms existing state-of-the-art methods in terms of both visual quality and computational efficiency. The project aims to further refine this approach and explore its applications in various rendering scenarios.



References

- [LZZY18] Yu Liu, Changwen Zheng, Quan Zheng, and Hongliang Yuan. Removing monte carlo noise using a sobel operator and a guided image filter. *Vis. Comput.*, 34(4):589–601, apr 2018.